Proportional Relationships: Four Representations
Answer Key

Grades 6–7 Mathematics

Quick Answers

1. 4
2. 2
3. (a) the equation = 4.5, (b) y when x = 10 = 45
4. table entry at 1 hour = 40, table entry at 2 hours = 80, table entry at 3 hours = 120, —
5. (a) cost of 6 kg = 21, —
6. (a) the equation = 2.5, (b) x when y = 20 = 8
7. (a) the equation = 1.5, (b) distance in 10 seconds = 15
8. (b) pages in 2.5 minutes = 30, —
9. (a) km in 1.5 hours = 90, —
10. (a) the equation = 0.75, (b) servings from 15 cups = 20
11. 0.5
12. (a) equation for Tank A = 4.5, (b) rate comparison = <, (c) extra liters after 8 minutes = 4

What is verified

8 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 4, 5, 8, 9. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 6–7 Mathematics
7.RP.A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 24 tagged checks.

1. Warm-up: equation from the graph.

Solution: The line passes through the origin and the marked point $(2, 8)$, so $k = \dfrac{8}{2} = 4$. The equation is $\boxed{y = 4x}$

2. Warm-up: unit rate for apples.

Solution: The line passes through $(0, 0)$ and the marked point $(5, 10)$, so the rate is $\dfrac{10}{5}$ — $\boxed{2}$ dollars per pound.

3. Table $(2, 9)$, $(4, 18)$, $(6, 27)$: equation, then predict.

Solution (a): Every row gives the same ratio: $\frac{9}{2} = \frac{18}{4} = \frac{27}{6} = 4.5$, so $k = 4.5$ and the equation is $\boxed{y = 4.5x}$

Solution (b): Substitute $x = 10$: $y = 4.5 \cdot 10$, so $\boxed{45}$

4. Tour bus at 40 km per hour: table and graph.

Solution (a): Distance $= 40\times$ hours: at 1 hour $\boxed{40}$ km; at 2 hours $40 \cdot 2$, giving $\boxed{80}$ km; at 3 hours $40 \cdot 3$, giving $\boxed{120}$ km.

Solution (b): *(instructor-judged)* The points $(1, 40)$, $(2, 80)$, $(3, 120)$ lie on one straight line through the origin — the graph should show that line.

5. Rice at $y = 3.5x$ dollars for x kg.

Solution (a): $y = 3.5 \cdot 6$, so 6 kg costs $\boxed{21}$ dollars.

Solution (b): *(instructor-judged)* The point $(1, 3.5)$ says that 1 kilogram costs \$3.50 — it is the unit price, the value of k read straight off the graph.

6. Table $(3, 7.5)$, $(5, 12.5)$: equation, then solve.

Solution (a): $k = \frac{12.5 - 7.5}{5 - 3} = \frac{5}{2} = 2.5$ (or $\frac{7.5}{3} = 2.5$ from one row). The equation is $\boxed{y = 2.5x}$

Solution (b): Solve $2.5x = 20$: $x = \frac{20}{2.5}$, so $\boxed{8}$

7. Hiker's graph through $(4, 6)$: equation, then distance for 10 hours.

Solution (a): $k = \frac{6}{4} = 1.5$, so the equation is $\boxed{y = 1.5x}$

Solution (b): $y = 1.5 \cdot 10$, so the hiker walks $\boxed{15}$ km.

8. Printer at 12 pages per minute: sketch, then read off 2.5 minutes.

Solution (a): *(instructor-judged)* The graph is the straight line through the origin with slope 12 — through $(1, 12)$, $(2, 24)$, $(3, 36)$.

Solution (b): Pages $= 12 \cdot 2.5$, so $\boxed{30}$ pages print.

9. Train at $y = 60x$: distance, then interpret k.

Solution (a): $y = 60 \cdot 1.5$, so the train covers $\boxed{90}$ km.

Solution (b): *(instructor-judged)* The 60 is the unit rate: the train travels 60 kilometers in each hour — its speed.

10. Soup table $(4, 3)$, $(8, 6)$ servings-to-cups: equation, then invert.

Solution (a): $k = \frac{3}{4} = 0.75$ cups per serving, so the equation is $\boxed{y = 0.75x}$

Solution (b): Solve $0.75x = 15$: $x = \frac{15}{0.75}$, so $\boxed{20}$ servings.

11. Plant A through $(2, 9)$ vs. Plant B at $y = 4x$: rate difference.

Solution: Plant A's rate: $\frac{9}{2} = 4.5$ cm per week. Plant B's rate: 4 cm per week. Difference: $4.5 - 4$, so Plant A grows $\boxed{0.5}$ cm more per week.

12. Challenge — Tank A (graph through $(4, 18)$) vs. Tank B (table).

Solution (a): $k = \frac{18}{4} = 4.5$ liters per minute, so Tank A's equation is $\boxed{y = 4.5x}$

Solution (b): Tank B's rate from the table: $\frac{10}{2} = 5$ liters per minute (check: $\frac{30}{6} = 5$). Comparing rates: $\boxed{4.5 < 5}$ — Tank B fills faster.

Solution (c): After 8 minutes: Tank B holds $5 \cdot 8 = 40$ liters, Tank A holds $4.5 \cdot 8 = 36$ liters. The faster tank holds $40 - 36$, that is $\boxed{4}$ more liters.